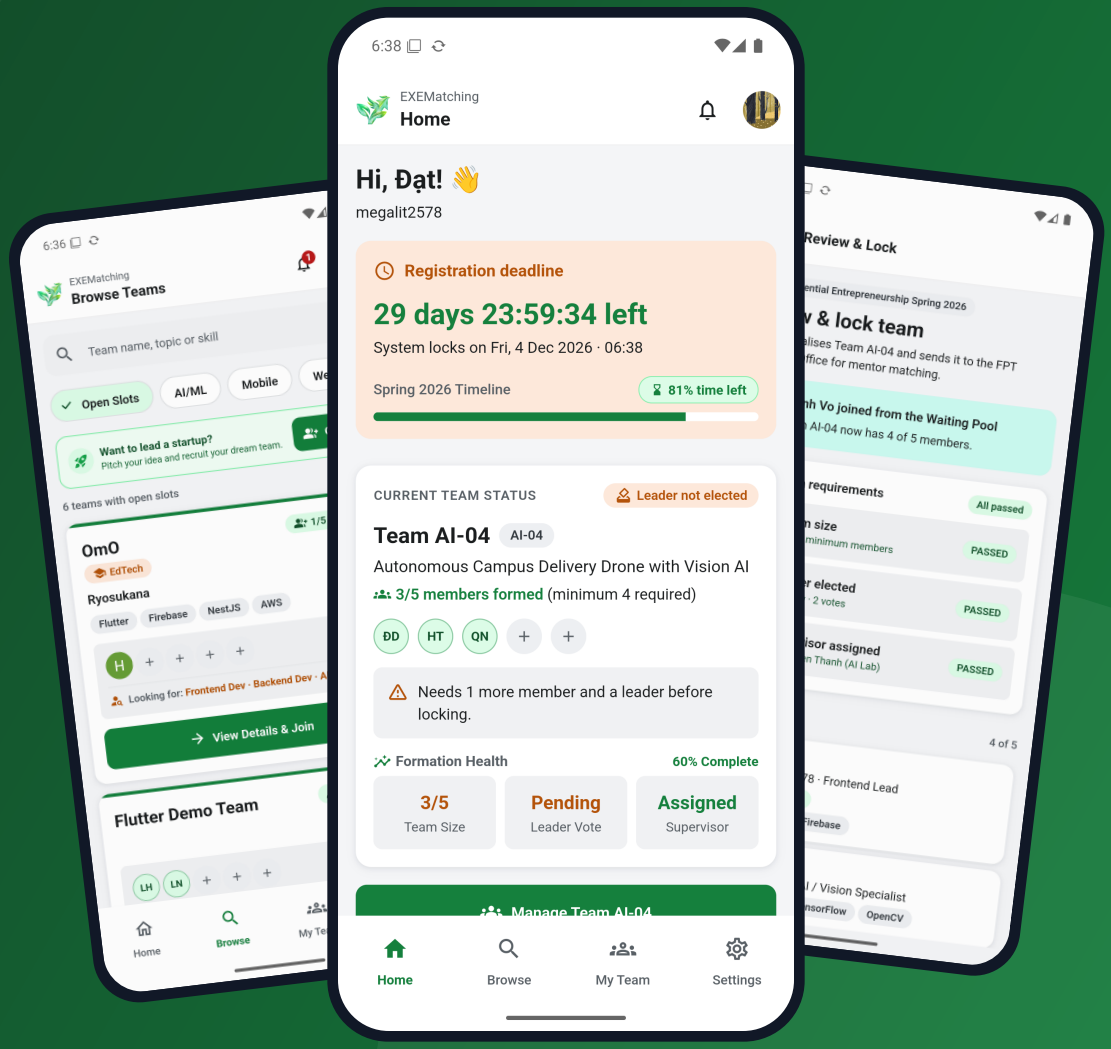
EXEMatching

PRM393 · MOBILE PROGRAMMING · LAB 3

Find your capstone team.
Elect a leader.
Lock the roster.

A cloud-enabled Flutter app on Firebase, tested end to end on a real emulator by Google ARTEMIS.

Flutter · MVVM | 7 Firebase services | 13 E2E cases × 3 runs | Gemini code review



Team DEVORGANIZE222 · Firebase project prm393-lab-3-exematching

LAB 2 RECAP

Who we built it for



Minh Lê

Final-year Software Engineering student

Persona · Lab 2

> *I have two weeks to register a capstone team. I don't know who has the skills we need, nobody wants to say who should lead, and if one person drops out we can't lock.*

GOALS

- Teammates with matching skills
- A fair leader vote
- Locked before the deadline

THREE PAINS, THREE FLOWS

Finding teammates with the right skills

Browse open teams by topic, skill and open slot, and join one in two taps.

Choosing a leader without peer pressure

Votes stay hidden until you vote; a majority elects the leader.

Recovering when a member drops out

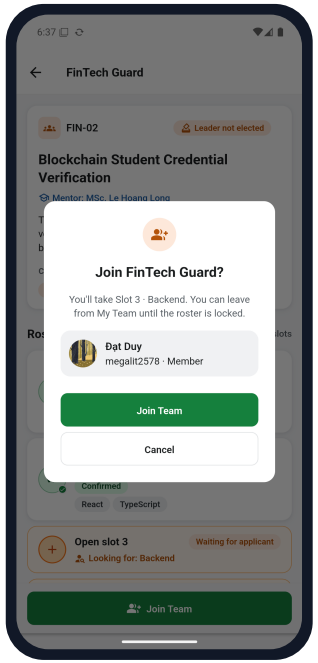
A team below 4 members can invite students from the Waiting Pool, then lock.

Success = a locked team of 4–5 members with an elected leader, before the deadline.

THE 3 CRITICAL FLOWS

From browsing to a locked roster

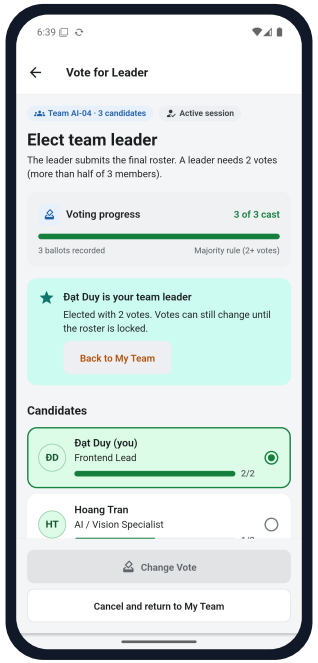
1 Discover & join



Home → Browse → Team detail → Join

Alt: team fills first · no results · offline

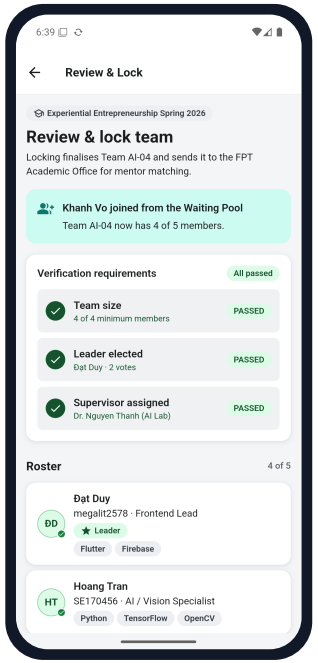
2 Elect a leader



My Team → Vote → majority wins

Alt: tallies hidden until you vote · change vote

3 Lock the roster

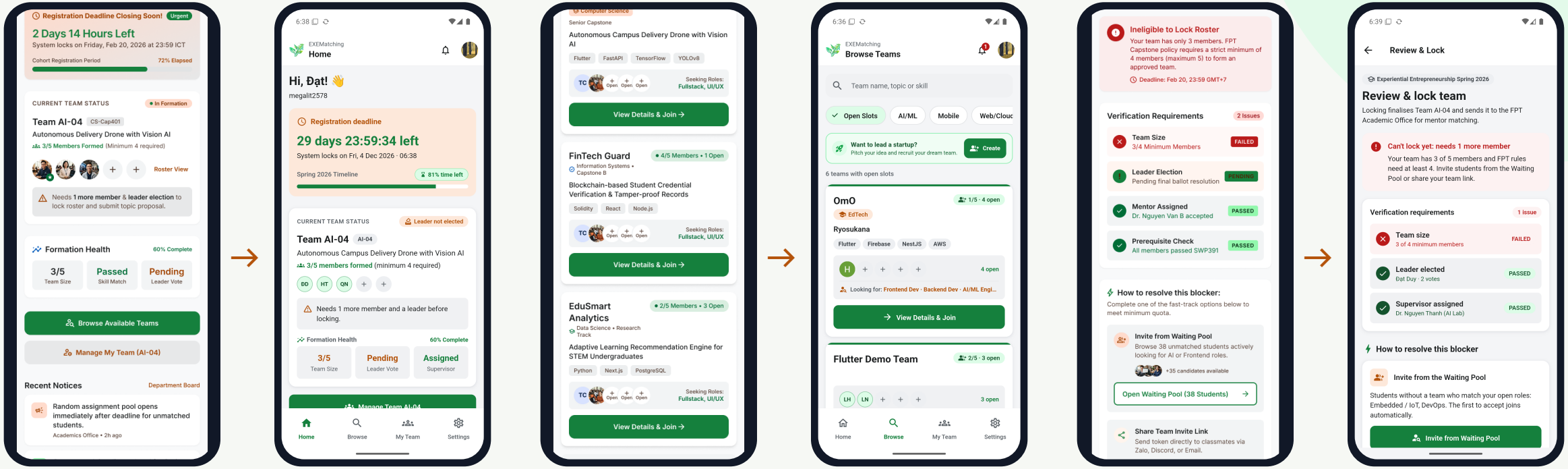


Review & Lock → Waiting Pool → Lock

Alt: fewer than 4 members blocks the lock

DESIGN TRACEABILITY

Figma frames, built in Flutter



Figma → App

SCR_02 · Home

Figma → App

SCR_03 · Browse

Figma → App

SCR_07 · Review & Lock

Deliberate deviations

- DV-01 "Team" everywhere
- DV-04 Direct join
- DV-08 Bell + Settings tab
- DV-09/10 WCAG AA contrast
- DV-14 Icon + label, not colour only

16 in docs/design-traceability.md

ARCHITECTURE

MVVM with one-way dependencies

Screens & widgets lib/screens · lib/widgets

Declarative UI only: no Firebase SDK, no business rules

↓

ViewModels lib/viewmodels

ChangeNotifier owning ViewState<T>, validation and analytics calls

↓

Service interfaces lib/services

Contracts the view-models depend on; local fakes for tests

↓

Firebase implementations lib/firebase

Auth, Firestore, Storage, FCM, Analytics, Crashlytics, Remote Config

Every screen renders one ViewState<T>

Loading | Empty + action | Failure + retry | Data

0

Firebase imports in lib/screens + lib/widgets

Counted from the source when this deck is generated.

One composition root

lib/app/providers.dart wires the Firebase services into the view-models with Provider.

CLOUD BACKEND

7 Firebase services, each with console proof

VERIFIED

Authentication

Google Sign-In, limited to the class roster

Users tab · roster rejection (TC-01)

VERIFIED

Firestore

Teams, votes and the Waiting Pool in transactions

Team documents · deployed rules

VERIFIED

Storage

Team PDF uploaded to reports/{uid}/

Uploaded PDFs · owner-only rules

VERIFIED

Cloud Messaging

Invites and updates in all 3 app states

Foreground · background · terminated

VERIFIED

Analytics

10 events, IDs and counts only

DebugView · Logcat per E2E run

VERIFIED

Crashlytics

Crashes and non-fatals with user ID and keys

Test crash + non-fatal, Nov 4

VERIFIED

Remote Config

Registration and pool switches, live

Published, seen live, reverted

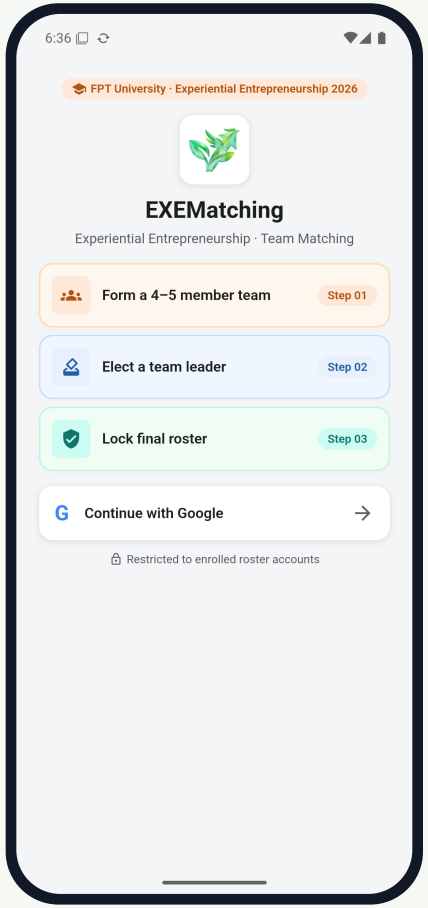
Security rules

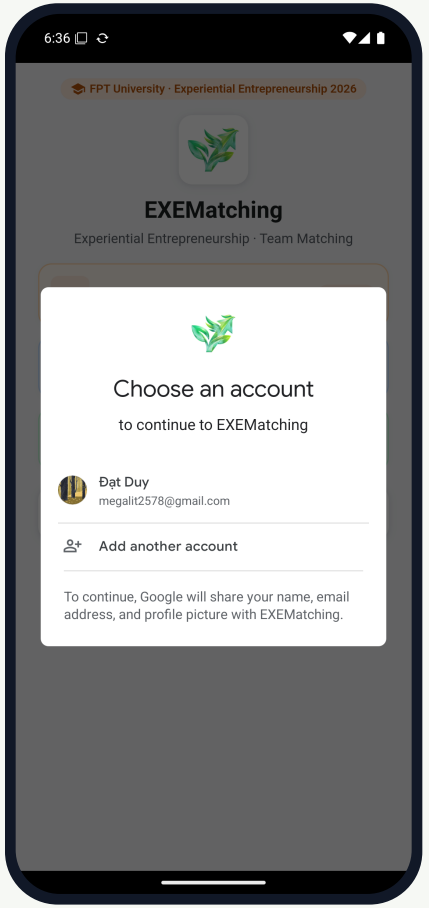
Firestore and Storage rules, deployed and screenshotted from the console.

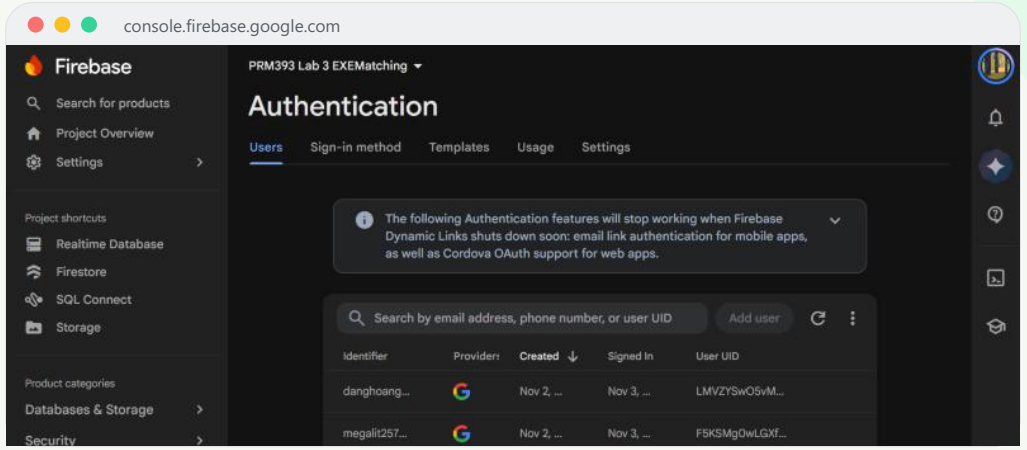
VERIFIED means the console or device screenshots are in evidence/ when this deck is generated.

CLOUD SERVICE 1 · AUTHENTICATION

Google Sign-In, roster only







Authentication → Users: Google-provider accounts with their UIDs

Google credential → FirebaseAuth

authStateChanges() drives routing; sign-out clears Google and Firebase.

Roster check in app and rules

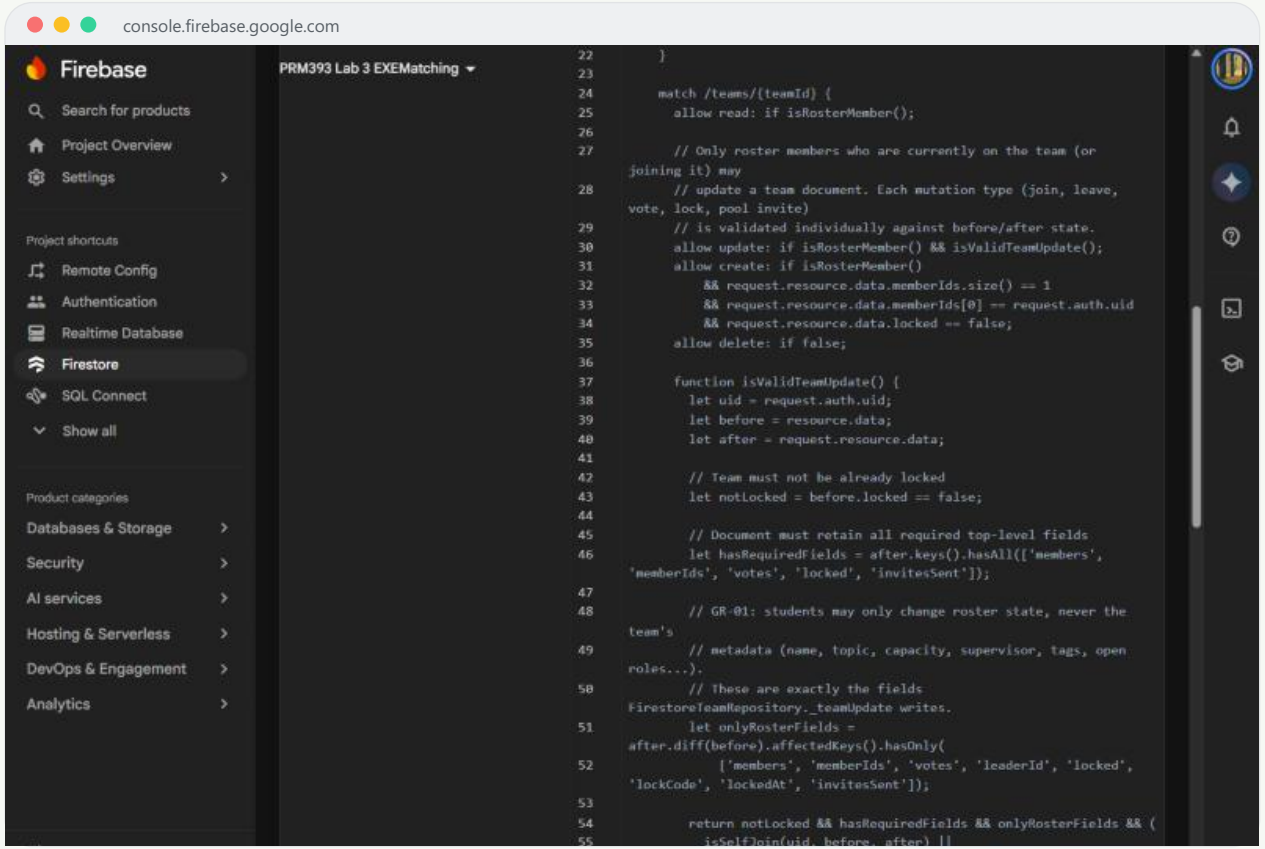
allowedEmails/{email} must be enabled; rules check isRosterMember().

Not on the roster? Rejected

E2E TC-01 passed 3/3 runs.

CLOUD SERVICE 2 · CLOUD FIRESTORE

Transactions and tight security rules



Deployed rules: a student may change only roster fields (GR-01)

5
mutations run in runTransaction

8
fields a student may change

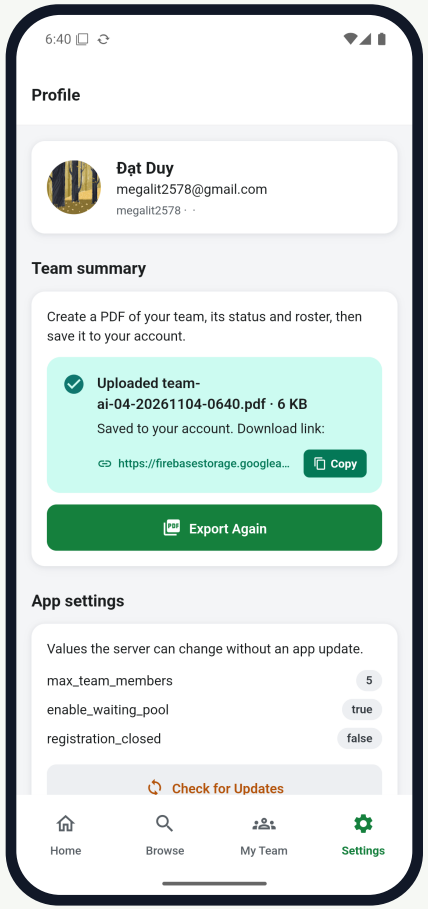
Join, leave, vote, lock, invite
Each re-reads the team inside the transaction, so two students can't take the last slot.

affectedKeys().hasOnly([...])
Topic, capacity and supervisor can't be edited by a client.

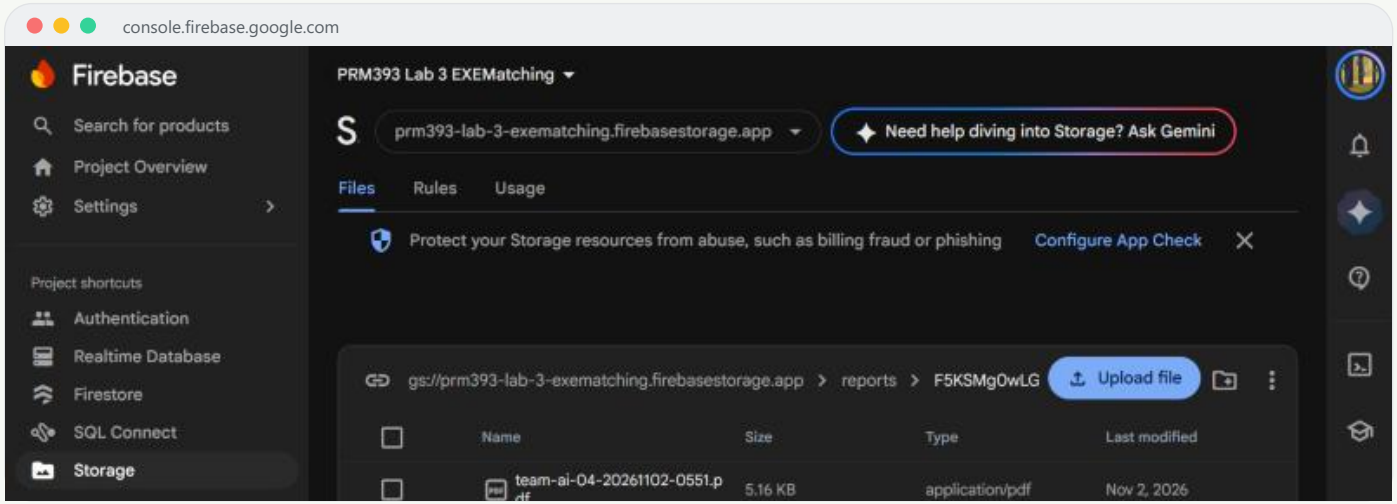
Waiting Pool privacy
Leaders read only unassigned candidates; assigned records stay private.

CLOUD SERVICE 3 · CLOUD STORAGE

Team PDF, private to its owner



Export & Upload



Storage → reports/{uid}/: PDFs uploaded by the app (E2E TC-13)

firebase/storage.rules

```
match /reports/{userId}/{allPaths=**} {
  allow read, write: if request.auth != null
                     && request.auth.uid == userId;
}
```

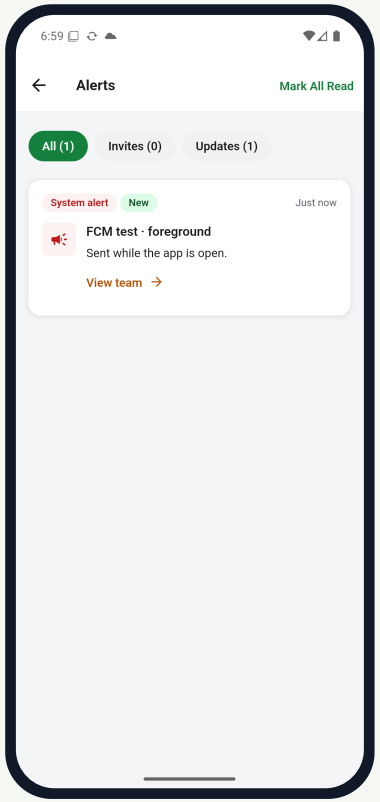
PDF built on the device

Upload progress 0 → 100 %

Download link shown

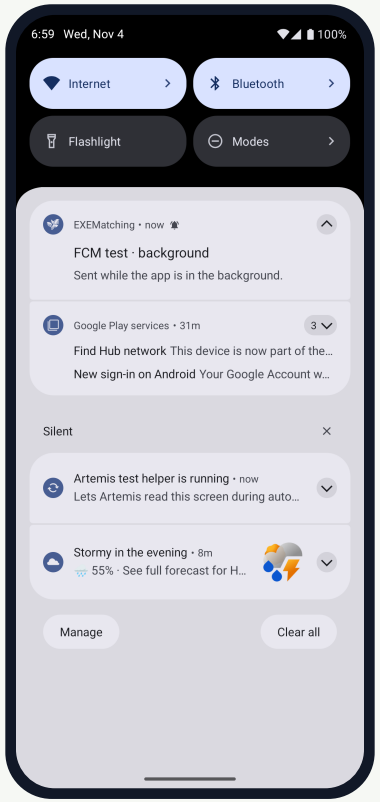
One test message, three app states

Foreground



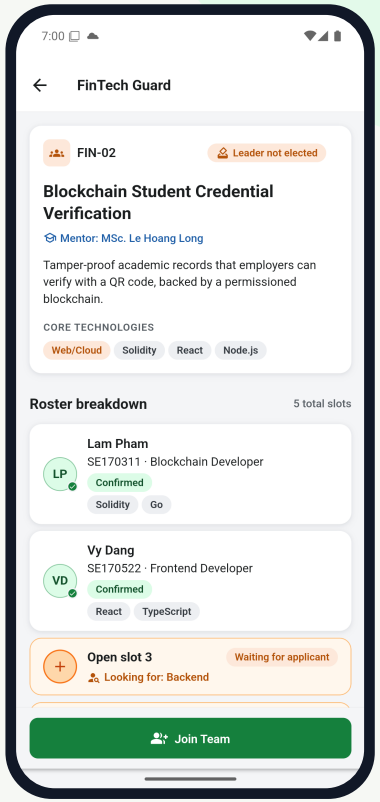
onMessage stores it; it appears in Alerts at once

Background



Android shows it; a tap runs onMessageOpenedApp

Terminated



Process killed; a tap cold-starts the app into the team

CLOUD SERVICE 5 · GOOGLE ANALYTICS

10 events, no personal data

Recommended events

- login
- logout
- search_teams
- view_team

Domain events

- create_team
- join_team
- cast_vote
- invite_pool
- lock_roster
- export_report

IDs and counts only

search_teams sends query_length, never the search text; no email or name.

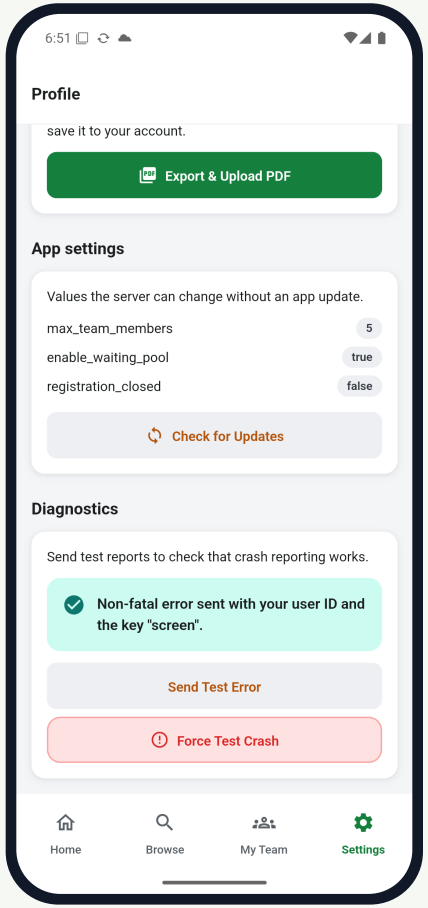
User ID after sign-in

setUserId(uid) links events to the account, not to a name.

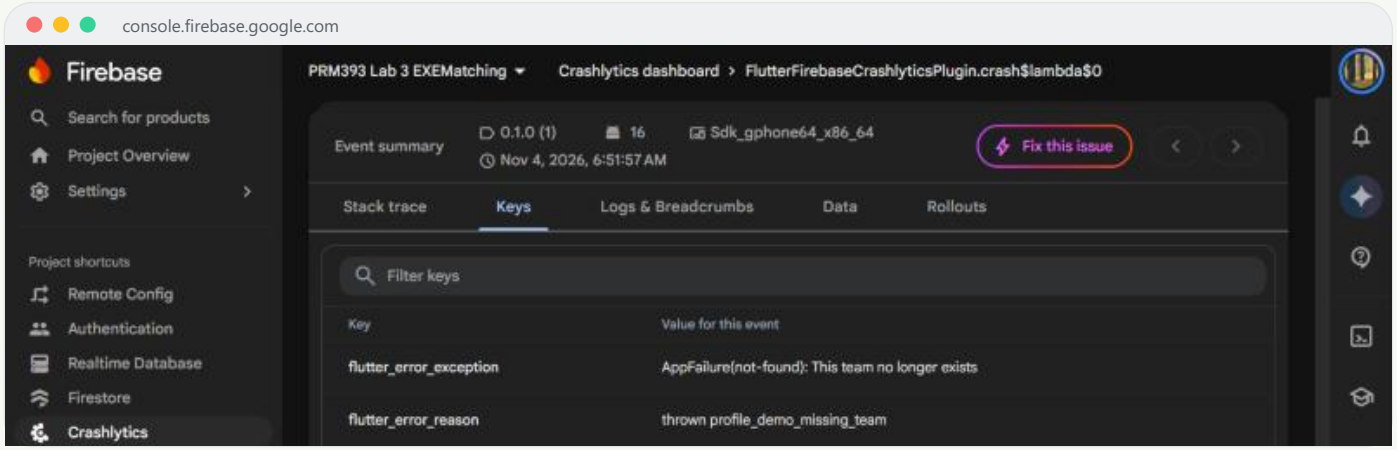
Checked twice

DebugView screenshots, and every E2E run saves the events it caused from Logcat.

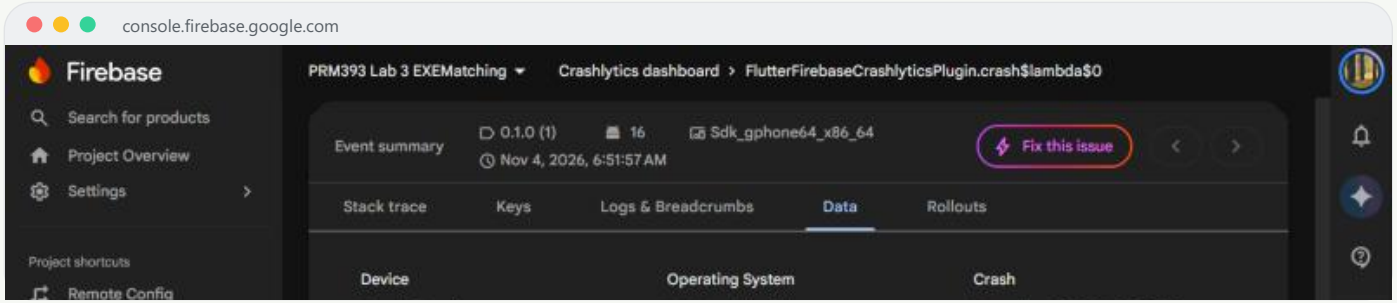
Crashes and handled errors, with context



Send Test Error



Test crash of Nov 4, 06:51:57 · Keys: screen = profile



Same event · Data: the user ID is the Firebase Auth UID

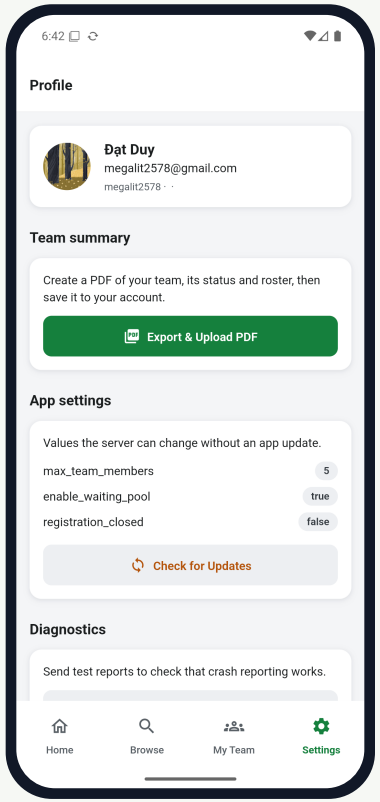
FlutterError.onError → fatal

recordError → non-fatal

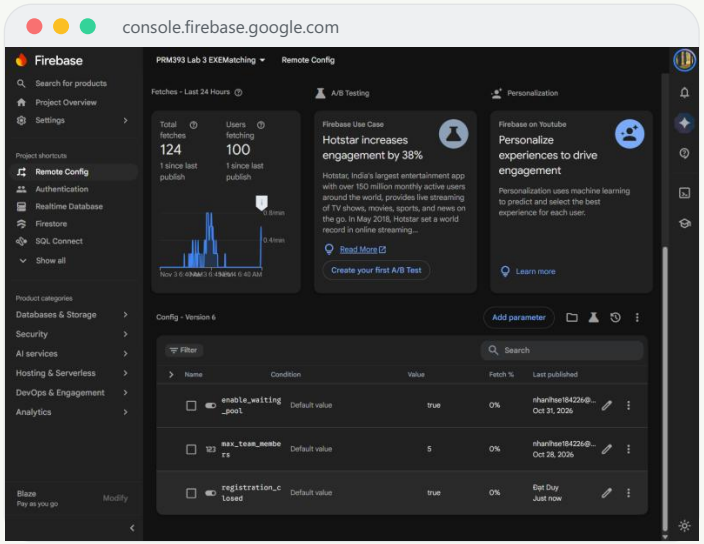
keys: screen · student_cohort · team_id

CLOUD SERVICE 7 · REMOTE CONFIG

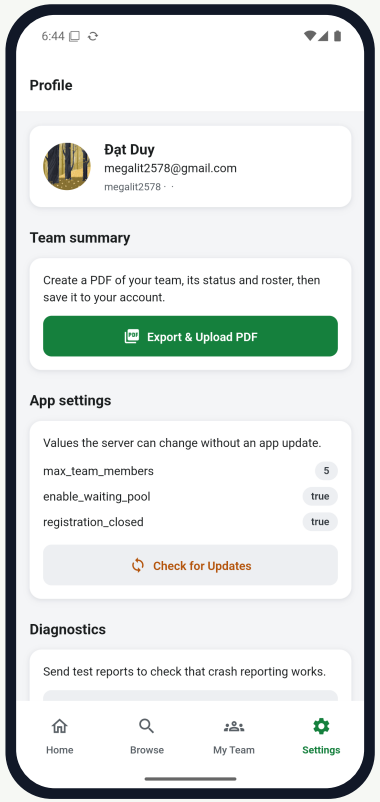
Close registration without an app update



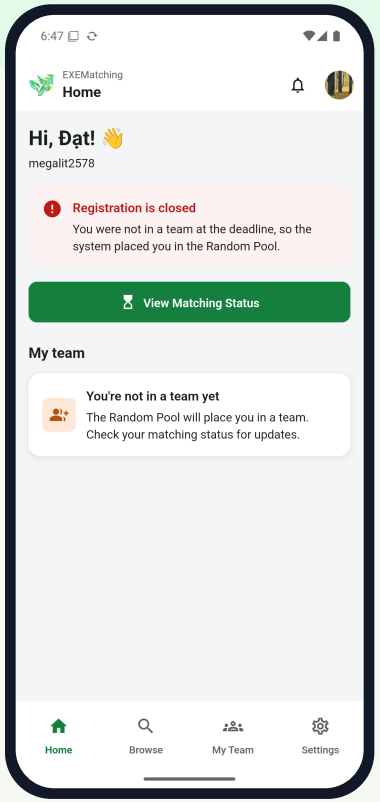
Before · false



registration_closed = true, published



After · true



Home: closed

onConfigUpdated activates the new value and the view-models rebuild: the open app changed within a minute, with no restart or reinstall. The parameter was deleted again afterwards.

QUALITY · GOOGLE ARTEMIS

13 cases × 3 runs on a real emulator

24 / 39 runs passed in the final run (e2e_test/evidence/junit.xml)

Every attempt, as recorded



The agent's SUCCESS is not trusted

Each run is checked against the accessibility tree, Logcat and Firestore.

Failed runs are explained

Bugs found, harness fixes and Gemini free-tier 429s, per trace ID: docs/test-plan.md §3.

Test case		Run 1 · 2 · 3
TC-01	Account not on roster	✓ ✓ ✓
TC-02	Sign in with roster account	✓ ✓ ✓
TC-03	Search & filter	✓ ✓ ✓
TC-04	Empty state + clear filters	✓ ✓ ✓
TC-05	Offline error + retry	✓ ✓ ✓
TC-06	Join a team with an open slot	✓ ✓ ✓
TC-07	Team fills before confirming	✗ ✗ ✗
TC-08	Tallies hidden until voting	✗ ✗ ✗
TC-09	Vote elects the leader	✗ ✗ ✗
TC-10	Lock blocked → pool recovery	✗ ✗ ✗
TC-11	Share link → Android share sheet	✗ ✗ ✗
TC-12	Decline invite + Undo	✓ ✓ ✓
TC-13	Export & upload team PDF	✓ ✓ ✓

AI-ASSISTED CODE REVIEW

Gemini reviews the PR, people decide

Pull request → gemini_review.py · one API request → gemini-3.5-flash-lite → Inline review comments

GR-01	Security · High	isSelfJoin only checked the roster size, so a client could also rewrite topic, capacity or supervisor.	ACCEPTED — Fixed for all 5 team mutations · c9543a3
GR-02	Concurrency · Med	Querying the Waiting Pool outside the transaction could let two teams pick the same student.	REJECTED — The client SDK can't query in a transaction; the candidate is re-read inside it
GR-03	Reliability · Med	A missing Firebase CLI login raised a bare KeyError in the E2E backend.	ACCEPTED — Actionable message · ee6ec78
GR-04	Reliability · Low	The Remote Config start-up catch dropped the error message.	ACCEPTED — Logged, defaults kept · d6926d1
GR-05	Reliability · Med	Slicing the error body to 2000 chars might split a UTF-8 character.	REJECTED — False positive: decoded before slicing

WHAT TESTING TAUGHT US

Real bugs, found on a real device

Bugs the runs exposed, all fixed

Invite from Waiting Pool always failed

Rules hid unassigned pool records from leaders (PERMISSION_DENIED).

Typing did nothing for TalkBack and ARTEMIS

A Semantics wrapper created a second, non-editable text field.

Voting showed a false error

cast_vote sent a bool parameter that Analytics rejects.

Undo after declining never appeared

The card was disposed before the snackbar showed.

Lessons

An agent's "success" is not evidence

TC-12 reported the invite restored while the screen showed none.

Only recorded runs count

Trace IDs, screenshots and Firestore checks, never a typed-in 39/39.

Test rules with the real flows

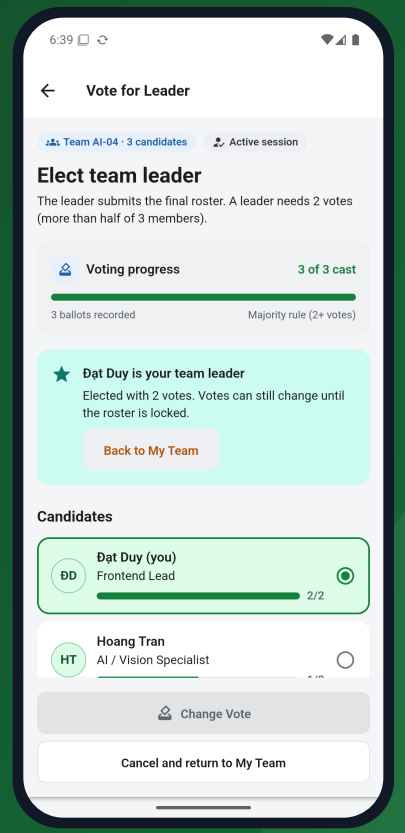
A correct-looking rule broke Flow 3 until the app ran against it.

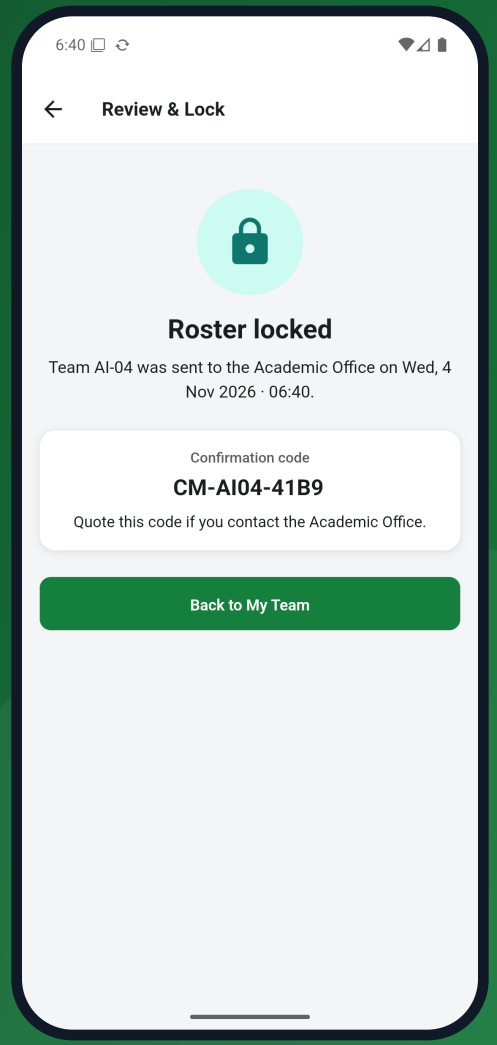
ARTEMIS Pro exploratory session: not run yet (docs/test-plan.md §4).

THANK YOU

Questions?

- github.com/DEVORGANIZE222/Lab3_Cloud-Enabled_Mobile_App
- docs/test-plan.md · evidence/ · report/Lab03_DEVORGANIZE222.pdf
- Firebase project prm393-lab-3-exematching





Team DEVORGANIZE222 · PRM393 Mobile Programming · FPT University